

AARTHI RAMALINGAM

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SUMMARY

Machine learning engineer with 6 years of expertise in designing and implementing impactful AI driven solutions using AWS. Achieved and enhanced customer experience by automating generation of product description, achieved 40% improvement in OCR accuracy and enhanced search functionality using AWS services. Improved project performance by 30% through clustering and BERT based topic modeling. Optimized data pipelines for better scalability and efficiency. Experienced in mentoring, agile methodologies.

PROFESSIONAL EXPERIENCE

HEXAWARE TECHNOLOGIES

BANGALORE

Senior Machine Learning Engineer

2022-Present

- Engineered an advanced query-response system utilizing Azure OpenAI and ChromaDB that processes text-rich PDFs; employed embedding methods for similarity search(RAG), boosting user satisfaction with accurate responses by over 30%.
- Pioneered and implemented advanced machine learning models using AWS Bedrock (LLM) to generate product stories from user images and file text through prompt engineering. Improved data integration and storage with AWS S3 via API Gateway, resulting in increased customer satisfaction, reduced costs, and scalable solutions for large data volumes
- Optimized and enhanced Optical Character Recognition (OCR) Techniques for medical document extraction with AWS Comprehend and Textract(text analysis), increasing data extraction processing accuracy by 15% and improving workflows for 15+ medical staff members
- Integrated custom entity recognition algorithms into existing workflows, resulting in the identification of over 1,000 unique queries monthly; findings to fix the three biggest causes of irrelevant results improved overall user satisfaction ratings.
- Led a series of hands-on workshops for junior team members, imparting machine learning and data science strategies that streamlined project timelines by 25%, enhancing overall technical proficiency within the group.
- Communicate closely with cross-functional teams during sprint cycles implementing agile methodologies resulting in streamlined workflows that increased task completion rate efficiency by nearly one-third compared to prior sprints

TATA CONSULTANCY SERVICES

PUNE

Machine Learning Engineer

2018-2022

- Developed a ML model for ticket classification, reducing misrouted incidents by 30% and enhancing operational efficiency through automated processing and AI-driven solutions
- Engineered a topic modeling solution using BERT(transformer) Embeddings for analyzing customer reviews, including data cleaning, data munging, data visualization, data extraction, preprocessing, and feature transformation,data transformation to derive actionable insights based on user requirements
- Constructed and deployed a text classification model for healthcare applications, optimizing data preprocessing and Embeddings to improve accuracy in injury type categorization
- Created a resume parsing solution utilizing Natural Language Processing (NLP) and Named Entity Recognition (NER) techniques to extract candidate details, integrating preprocessing and feature extraction within a Flask application
- Applied statistics analysis(coding) and feature engineering to enhance model performance and scalability, leading to improved accuracy and efficiency in deployed solutions

SKILLS

- Programming Language: Python | Frameworks: Tensorflow, Keras, Langchain

- Machine Learning Algorithms: Linear, logistics regression, Decision Trees, Random Forest, XGBOOST
- Cloud: AWS (S3, lambda, IAM, Database, CloudWatch, SQS, Bedrock, Guardrails, ECS, CI/CD)
- Library: Scikit Learn | Database: Vector database, SQL
- Tools: Git, Mlflow, Dagshub | Certifications: Certified AWS Machine Learning(2023)
- Domains: Artificial Intelligence, Machine Learning, Deep Learning, Retail, Healthcare, Insurance, NLP(NLTK)

EDUCATION

SASTRA UNIVERSITY

Bachelor of Engineering in Bioinformatics

Thanjavur, TamilNadu

2013-2017

HIGHER SECONDARY CERTIFICATE AND SECONDARY SCHOOL CERTIFICATE

Savitri vidyasala higher secondary school

Kamakoti vidyalaya

Trichy, TamilNadu

2011-2013

2011

MY PERSONAL PROJECTS

- Implemented project using mlflow and dagshub.